HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 227: Data Communication

Batch: 27th
Date: 20/06/2026

Mid-Term Examination

Semester: Spring 2026
Time:1 Hour 30 Minutes
Total Marks : 30

Answer any five questions: $5 \times 6 = 30$

		CLOs	Marks
1.	(a) Explain the five components of a data communication system.	CLO1	3
	(b) What are the differences between simplex and half-duplex transmission?	CLO1	3
2.	(a) Name the four basic network topologies, and cite an advantage of each type.	CLO1	3
	(b) Write the differences between the duties of the IETF and IRTF.	CLO1	3
3.	(a) Write the reasons for the lack of OSI Model's success.	CLO1	3
	(b) Explain Multiplexing and De-multiplexing in the TCP/IP protocol suite.	CLO5	3
4.	(a) A file contains 2 million bytes. How long does it take to download this file using a 56-Kbps channel and 1-Mbps channel?	CLO2	3
	(b) Describe Nyquist Bit Rate and Shannon Capacity.	CLO2	3
5.	(a) Name and explain three types of transmission impairment.	CLO4	3
	(b) Explain Amplitude, Frequency and Phase.	CLO3	3
6.	(a) Distinguish between a signal element and a data element.	CLO3	3
	(b) Define baseline wandering and DC component.	CLO1	3
7.	(a) Draw a Summary table for line coding schemes.	CLO2	3
	(b) Draw the graph of the NRZ-I scheme using each of the following data streams, assuming that the last signal level has been positive. I.00110011 II.11111111 III.01010101	CLO2	3

HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: ENS 221		**Course Title: Environmental Science**
Mid-Term Examination		Semester: Spring 2026
Date: 6/17/2026	Total Time: 1:30 Hours	**Total Marks: 30**

*** If necessary, draw relevant graphs, diagrams to support your explanations.:** $3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

Part A (Answer any One)

			CLOs	Marks
Q1.	(a)	Define Environment.	CLO1	[2]
	(b)	Differentiate between environmentalist and scientists with examples.	CLO2	[4]
	(c)	How have human activities, such as urbanization and industrialization, led to environmental degradation?	CLO2 CLO3	[4]
Q2.	(a)	What is the main function of ecosystem?	CLO1	[2]
	(b)	"Environmental Science is a multidisciplinary approach" Explain it.	CLO2	[4]
	(c)	Discuss the abiotic and biotic components of an ecosystem.	CLO3	[4]

Part B (Answer any Two)

			CLOs	Marks
Q3.	(a)	What is the water cycle?	CLO1	[1]
	(b)	What are different forms of precipitation?	CLO2	[2]
	(c)	Explain the process of the hydrological cycle with its importance.	CLO3	[4]
	(d)	What strategies can be adopted to ensure water security?	CLO2	[3]
Q4.	(a)	Define the greenhouse effect.	CLO1	[1]
	(b)	Write the contributes to global warming and climate change due to greenhouse effect.	CLO2	[4]
	(c)	Draw and explain the layers of the atmosphere.	CLO3	[5]
Q5.	(a)	What is sustainable agriculture?	CLO1	[1]
	(b)	Describe its key principles and why it is important for both food security and the environment.	CLO1 CLO2	[3]
	(c)	List the five main indicators of food security and explain how climate change can affect food security.	CLO1 CLO3	[6]

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE221: Computer Algorithms

Batch: 27th	**Mid-Term Examination**	Time: 1 Hour 30 Minutes
Date: 08-06-2026	Semester: Spring 2026	Total Marks: 30

Answer any five questions of the followings. $5 \times 6 = 30$

			CLOs	Marks
1.	(a)	What is the difference between best-case and worst-case complexity?	CLO1	2
	(b)	Analyze the complexity from the following recurrence: $T(n) = \begin{cases} T(n/2) + 1 & if\ n > 1 \\ 1 & otherwise \end{cases}$	CLO1	4
2.	(a)	What is the worst-case time complexity of Merge Sort, and when does this happen?	CLO2	2
	(b)	Apply Merge Sort Algorithm to sort the elements of the array: A[9]: 36, 25, 37, 23, 42, 14, 47, 53, 35	CLO3	4
3.	(a)	Can recursion always be converted to iteration? Give an example.	CLO2	2
	(b)	Let, you have two strings A= "ICT" and B= "BCT". Now calculate their longest common subsequence using recursion.	CLO2	4
4.	(a)	Is it possible for multiple optimal solutions to exist? When?	CLO1	1
	(b)	Suppose you have 10 elements in an array as follows: (see table below) Draw the recursion tree to find the maximum and minimum elements from the array using the recursive **Max-Min Algorithm**.	CLO3	5

Address	1	2	3	4	5	6	7	8	9	10
Elements	89	-8	35	23	85	-17	25	31	-20	12

			CLOs	Marks
5.	(a)	What is the role of sorting in greedy algorithms? Why is it often necessary?	CLO2	2
	(b)	You are given the following 8 intervals: A: (2,4), B: (3,5), C: (4, 9), D: (5, 8), E: (6, 7), F: (8, 11), G: (9, 13), H: (10, 12) Schedule the intervals using the Greedy Method.	CLO3	4
6.	(a)	A gardener wants to plant flowers in the largest empty space first versus divide the garden into quadrants, plant each, then combine visually. Which approach is greedy and which is divide and conquer?	CLO2	2
	(b)	Let you have four slots and six jobs with the following profits and deadlines respectively: $\{p_1, p_2, p_3, p_4, p_5, p_6\} = \{20, 17, 12, 10, 9, 8\}$ *and* $\{d_1, d_2, d_3, d_4, d_5, d_6\} = \{3, 1, 4, 1, 3, 2\}$ If each job takes one unit of time to execute, find the optimal solution using the Greedy Algorithm.	CLO3	4

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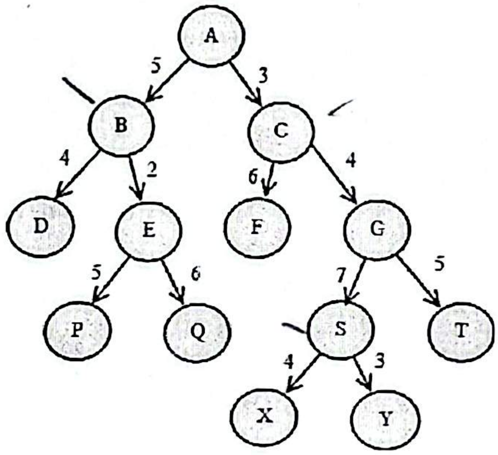
In Pursuit of Knowledge

7.	(a)	When is Prim's algorithm preferred over Kruskal's algorithm?	CLO4	2
	(b)	Consider the tolerance level of the following network is 12. Now split the network with the help of Greedy Method.	CLO3	4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 223: Object Oriented Programming

Batch: 27th		Semester: Spring 2026
Section: 1		Time: 1 Hour 30 Minutes
Date: 10/06/2026	**Mid-Term Examination**	Total Marks: 30

Answer any five questions from the following. $5 \times 6 = 30$

		Question	CLOs	Marks
1	(a).	Briefly explain **Object and Class** of Object Oriented Programming (OOP).	CLO1	3
	(b).	Explain the basic characteristics of OOP.	CLO1	3
2	(a).	Illustrate the block diagram of Java Virtual Machine (JVM). Briefly explain class loader of the JVM.	CLO3	3
	(b).	Write the differences between Object Oriented Programming and Structured Programming Language.	CLO1	3
3	(a).	*Explain method overloading in OOP.*	CLO1	3
	(b).	Write a Java method that finds the maximum and minimum value in an integer array.	CLO3	3
4	(a).	Define a constructor in OOP.	CLO1	1
	(b).	Why is 'this' keyword used in Java?	CLO3	2
	(c).	Write a program in Java to find if a number is prime or not using parameterized constructor.	CLO3	3
5	(a).	Print the string **'We love our country'** 10 times by using a for-each loop to iterate over an array of 10 elements.	CLO2	2
	(b).	Write a program in Java for Linear search. Sample I/O: Input: a = [1, 2, 3, 5, 7], x = 3 Output = Element found at index: 2 Input a = [1, 2, 3, 5, 7] x = 8 Output = -1	CLO2	4
6	(a).	Write the basic syntax of while and do…while loop in Java.	CLO3	2
	(b).	Write a Java program that accepts a student's numerical score (0 to 100) and uses a switch case statement to determine and print their corresponding letter grade based on the standard university scale below:	CLO2	4

Score Range	Grade
80 to 100	A+
75 to 79	A
70 to 74	A-

7 (a).	Explain Boolean and Float data types with examples.	CLO1	3
(b).	Consider the following code segment:	CLO2	3

```
public int countElements(int[] data) {
    int count = 0;
    for (int i = 0; i <= data.length; i++) {
        if (data[i] > 10); {
            count = count + 1;
        }
    }
    return count;
}
```

Identify the errors present in this code segment and rewrite the code segment so that it functions correctly.

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE225: Digital Logic Design

Batch: 27^{th} — Semester: Spring 2026

Date: 13/06/2026 — **Midterm Examination** — Time: 1 Hour 30 minutes

Total Marks: 30

Answer any five questions of the following. $5 \times 6 = 30$

1.	(a)	Find the 9's and 10's complements of the number $(76832)_{10}$.	CLO3	2
	(b)	Perform the following operations using complements showing all steps clearly. i. $(89345)_{10} - (45687)_{10}$ using 9's complement ii. $(111101)_2 - (01011)_2$ using 1's complement	CLO3	4
2.	(a)	Define the following terms: i. Canonical Form ii. Intermediate functions iii. Essential Prime Implicants	CLO1 CLO3	2 3 2
	(b)	Simplify the following function to the minimum numbers of literals. $F = XY + X'YW + XYZW + X'YW + X'ZW'$	CLO2	2 3
3.	(a)	Define Digital system. What are the advantages and disadvantages of Digital systems?	CLO5	3
	(b)	Derive POS from the following simplified SOP using a truth table. $F = Y'Z + X'Y$	CLO3	3
4.	(a)	Simplify the following function using a k-map: $F = \sum m\,(1, 2, 4, 6, 7, 9, 10, 14, 15) + d\,(5, 8, 11, 13)$	CLO3	3
	(b)	For the following intermediate function find out the Sum Of Minterms (SOM): $F = XY(X'+Z) + X(X+Y')(Y+Z')$	CLO3	3
5.		Given the following function: $F = \Pi M\,(4, 5, 6, 7)$ i. Find out the POS expression using truth table ii. Find the reduced POS iii. Draw the logic diagram of the reduced POS and calculate the cost of the circuit.	CLO3 & CLO5	6

$(\bar{x}_1 + x_2)(\bar{x}_1 + \bar{x}_2)$

SOP - into 0 +

POS - 1 into 0

6.	Consider the following function: $F = \sum m(0, 1, 2, 4, 5)$ i. Create the truth table and expression in SOP ii. Find the simplified SOP iii. Draw the logic diagram of the simplified SOP and calculate the cost of the circuit.	CLO3 & CLO5	6
7.	For the function $F(A, B, C, D) = \Sigma m(1, 2, 5, 6, 7, 8, 10, 15)$ i. Identify ALL prime implicants ii. Identify ALL essential prime implicants iii. If the above function had don't care conditions at m(3, 4, 13, 14), how would the prime implicants and essential prime implicants change?	CLO5	6

HAMDARD UNIVERSITY BANGLADESH

Computer system
CPU
control unit

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 229 : Computer Architecture

Batch: 27th

Time: 1 Hour 30 Minutes

Midterm Examination

Total Marks: 30

Answer any **five** questions. $5 \times 6 = 30$

1.	(a)	Distinguish between Computer Architecture and Computer Organization with examples.	CLO1	3
	(b)	Draw the hierarchical structure of a computer system showing Structure and Function at each level.	CLO1	3
2.	(a)	State the overflow rule with an example each.	CLO4	2
	(b)	(code below) What is the corresponding MIPS assembly code?	CLO4	4

```
void strcpy (char x[], char y[]) {
        int i; i = 0;
    while ((x[i] = y[i]) != '\0')
            i += 1;
}
```

What is the corresponding MIPS assembly code?

3.	(a)	A[3000] = h + A[3000]; Write down the MIPS Instruction and make a MIPS code table from it.	CLO4	3
				3
	(b)	Explain Booth's algorithm's principle and its advantages over standard multiplication.	CLO5	
4.	(a)	Simulate (-5) × (3) using Booth's algorithm with 4-bit representation. In	CLO4	3
	(b)	the following code segment, f, g, h, i, and j are variables. If the five variables f through j correspond to the five registers \$s0 through \$s4, what is the compiled MIPS code for this C if statement? And also draw its flowchart.	CLO4	3

```
if (i == j) {
f = g + h;
}
else {
f = g – h;}
```

0001
010
0110

5. (a) Draw a flowchart for Floating-point division operation. CLO3 2

(b) Draw the flowchart for floating-point addition/subtraction showing normalization steps. CLO3 4

6. (a) Draw the complete instruction cycle state diagram with interrupts. CLO3 2

(b) Here is a traditional loop in C: CLO3 4

```
while (save[i] == k)

 i += 1;
```

Assume that i and k correspond to registers `$S3` and `$S5`, and the base of the array save is in `$S6`. What is the MIPS assembly code corresponding to this C segment?

7. (a) Describe the underlying principles of hardware design CLO3 3

(b) CLO3 3

```
int leaf_example (int g, int h, int i, int j) {

int f;

f = (g + h) – (i + j);

return f;

}
```

What is the compiled MIPS assembly code for this function?